

Digital Radio Heralding Opportunities for R&D, Manufacturing and Exports

To provide enhanced audio quality, service reliability and additional features, countries around the world are adopting digital technology as the latest trend in radio, television and telecommunications. In keeping with the government's decision of transitioning towards the digital mode of transmission, All India Radio is switching from analogue to digital broadcasting in a phased manner with a target of complete digitisation by the year 2017



9950 AIR DRM 1800

DR R.C. CHOPRA

For complete digitisation, the technology adopted by All India Radio (AIR) is the digital radio Mondiale (DRM), which is an open standard having a number of operating modes that have been designed to utilise AM and FM broadcast bands. AIR is in the process of replacing and/or converting 72MW transmitters to digital ones all over the country, which will cover 70 per cent of the Indian population, representing around 800 million people.

Eight DRM transmitters have already been made operational by AIR. After a global tender, another 27 digital DRM transmitters have been ordered from which eight have been received and are under installation. Remaining

19 transmitters are expected to be received before April 2014. AIR already uses the technology for some of its international services.

The DRM standard and its salient features

DRM is the universal, standardised digital broadcasting system covering all broadcasting frequencies. The DRM standard comprises two major configurations.

The first is 'DRM30' which is intended for broadcasts on short, medium and long waves up to 30MHz. It provides large coverage areas and low power consumption.

The second is 'DRM+' which is the configuration for the VHF bands above 30MHz. It is tailored for local and regional coverage with broadcaster-

controlled transmissions.

The DRM system offers several benefits to the listeners, manufacturers, broadcasters and regulators. It provides improved audio quality, added data services, emergency warning alerts, targeted advertising and low power costs.

Another key differentiating factor is the display which is integral to the receiver. The display can be used for various applications including commercials. Some of the benefits of interest to the consumers are:

1. Availability of a wider range of services
2. Easier tuning and selection of programmes, for example, automatic switching between different transmitters
3. Improved formats such as stereo in the AM bands and surround sound in cars
4. Improved and more consistent sound quality
5. Programme-associated data, textual content description or even independent services such as traffic and weather information, news updates and, more importantly, disaster alerts.

Amongst the various data services offered is the DRM electronic programme guide (EPG), which allows listeners with appropriate receivers to access the broadcast schedule and set recording times accordingly, and Journaline—accompanying audio programmes with interactive textual information such as news or graphics.